GSE 518
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Problem Set 4

You must turn in problems: 1, 2, 4 and 5(b). However, you are strongly encouraged to work on all the problems before the midterm, since the material they cover will be part of the midterm. Please let me know if you have any questions.

1. Problem 2.24 in CB (a) and (b). (*Hint:* $\sum_{i=1}^{k} i$ and $\sum_{i=1}^{k} i^2$ are two well known series, where $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$ and $\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$.)

2. Let X be a random variable with pdf $f(x) = 2/x^3$ for $x > 1$, zero otherwise. Show:

   (a) $\int_{-\infty}^{\infty} f(x)\, dx = 1$
   (b) $E(X) = 2$
   (c) $Var(X)$ does not exist.

3. Let X be a random variable with mean μ and variance σ^2 such that $E[(X-\mu)^3]$ exists. The value of the ratio $E[(X-\mu)^3]/\sigma^3$ is often used as a measure of *skewness*, that is, as a measure of the asymmetry of a pdf/pmf about its mean (the skewness of a symmetric pdf/pmf is zero).

   (a) Show that $E[(X-\mu)^3] = E(X^3) - 3\mu E(X^2) + 2\mu^3$.
   (b) Calculate $E[(X-\mu)^3]/\sigma^3$ for the following distribution: $f(x) = 1/2$ for $-1 < x < 1$, zero otherwise.
   (c) Let X be a random variable with mean μ and variance σ^2 such that $E[(X-\mu)^4]$ exists. The value of the ratio $E[(X-\mu)^4]/\sigma^4$ is often used as a measure of *kurtosis*, that is, as a measure of how thick the tails of a pdf/pmf are (e.g., a pdf with high kurtosis infrequently takes on extremely large values, like the Cauchy distribution). Calculate it for the distribution in part (b).

4. Let X be a random variable which follows an uniform distribution with pdf $f(x) = 1/(b-a)$ for $x \in (a, b)$, zero otherwise ($b > a$). Derive the moment generating function of X and find the mean and variance of X. Can we use the mgf of a uniform-distributed random variable to derive its moments?

5. (a) *(Just for fun problem, i.e., you do not need to turn in this part of the problem. However, you do need to turn in part (b).)* Show that

   $$M_X(t) = e^{t\mu + \frac{t^2\sigma^2}{2}}$$

   is the moment generating function for a random variable X having a normal distribution with pdf

   $$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2},$$

where $-\infty < x < \infty$, $-\infty < \mu < \infty$ and $\sigma^2 > 0$. *(Hint:* This part is a little algebra intensive. Try to complete the square in the exponent of the exponential function to obtain the pdf of another normal distribution inside the integral and make this whole integral equal to 1 (since it is a pdf for some random variable). This approach, which is used a lot in probability to mathematically calculate expectations, is similar in nature to the one CB follows in Example 2.3.8 to derive the mgf of a Gamma distribution.).

(b) Use the moment generating function in (a) to find the mean and variance of X when it follows a normal distribution, $N(\mu, \sigma^2)$.